**Title: Automated Nutrient and pH Controller for Mycology & Food Security**

**Revision History:**

- **Version 1.0 (2025-08-24):** Initial Release
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- **License:** Open Source, MIT License

**1.0 Introduction** This document describes the design, implementation, and operation of an automated nutrient and pH controller for hydroponic and liquid culture systems. The system is built on the Arduino Due platform and is designed for applications in mycology and food security, ensuring optimal environmental conditions for cultivation. It features automated dosing, a robust pH calibration routine, and remote control via a Bluetooth interface.

**2.0 System Architecture** The system is composed of four main functional blocks:

- **Control Unit:** An Arduino Due, selected for its 3.3V logic, 12-bit ADC precision, and extended I/O capabilities.
- **Dosing Subsystem:** Four 12V DC peristaltic pumps, each controlled by a single FDS6612A N-Channel MOSFET acting as a low-side switch. This ensures efficient and precise liquid delivery.
- **Sensing Subsystem:** A pH probe and an associated 3.3V-compatible module. The system is calibrated using a two-point linear regression method for high accuracy.
- **User Interface:** A 2.2-inch TFT LCD for local display of system status and an HC-05 Bluetooth module for remote configuration and monitoring via a mobile device. Configuration data is stored on an SD card for persistence.

**3.0 Component List** | Component | Quantity | Description | | :--- | :--- | :--- | | Arduino Due | 1 | Microcontroller board | | FDS6612A MOSFET | 5 | N-Channel, logic-level MOSFET | | Peristaltic Pump | 4 | 12V DC, 400mA | | pH Sensor Module | 1 | 3.3V compatible | | HC-05 Module | 1 | Bluetooth Transceiver | | 2.2" TFT LCD | 1 | 220x176 with SD card cage | | Power Supply | 1 | 12V DC, >2A | | Breadboard | 1 | For prototyping | | Jumper Wires | Assorted | For connections | | 1N4001 Diode | 4 | Flyback diode for each pump motor |

**4.0 Hardware Connections 4.1 Peristaltic Pump Control**

- Connect Arduino digital pins 2, 3, 4, and 5 to the Gate of each MOSFET.
- Connect the Source of each MOSFET to the common ground of the circuit (Arduino GND and 12V power supply ground).
- Connect the Drain of each MOSFET to the negative terminal of a corresponding pump motor.
- Connect the positive terminal of each pump motor to the +12V rail of the power supply.
- Install a 1N4001 diode in reverse across each pump motor's terminals.

#### 4.2 pH Sensing

- Connect the pH sensor module's VCC to the Arduino's 3.3V pin and GND to GND.
- Connect the pH sensor module's analog output pin to the Arduino's A0 pin.

#### 4.3 User Interface

- Connect the TFT LCD module to the Arduino Due's headers. The module's built-in level shifters simplify this connection.
- Connect the HC-05 Bluetooth module's VCC to the Arduino's 5V pin, GND to GND, TX to Arduino RX1 (pin 19), and RX to Arduino TX1 (pin 18). Ensure a voltage divider is used on the TX to RX1 line, or use a bi-directional logic-level shifter.

**5.0 Software & Operation** The system is controlled by open-source firmware.

- **Dosing:** Automated dosing of nutrients (pumps 1-3) and pH-Up solution (pump 4) is triggered when pH is below the user-defined target.
- **Calibration:** The system requires a two-point calibration using pH 7 and pH 4 reference solutions. The `CALPH 7` and `CALPH 4` commands are sent via Bluetooth to perform this.
- **Configuration:** All system parameters are stored in the `Config` struct and are saved to a `ph_cfg.txt` file on the SD card using the `SAVE` command.
- **Monitoring:** The LCD provides live status, while the Bluetooth interface allows for remote monitoring and command execution.

#### 6.0 Maintenance & Troubleshooting

- **Maintenance:** Regularly check pH and nutrient solution levels. Clean the pH probe tip periodically to ensure accurate readings.
- **Troubleshooting:** If pumps fail to activate, check power supply connections and MOSFET wiring. If pH readings are erratic, re-run the two-point calibration.

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## A Smarter Way to Grow: The Sparkhope Hydroponics Controller

Welcome to the future of growing! 🌿 At SPARKHOPE.SPACE, we believe that everyone should have the tools to grow their own food, especially in a world where food security is more important than ever. That's why we're sharing this project: an open-source, automated hydroponics controller that takes the guesswork out of nutrient and pH management.

### What this Circuit Provides

Think of this circuit as your personal robotic gardener. It's designed to keep your liquid culture or hydroponic reservoir at the perfect pH level, 24/7. Here's what it does:

- **pH Monitoring:** A special probe constantly checks the pH of your nutrient solution, a bit like a digital thermometer for acidity.
- **Smart Dosing:** When the pH drops below your ideal level, the system automatically activates a tiny pump to add a small, precise dose of pH-Up solution. No more manual adjustments!
- **Automated Feeding:** It can also be programmed to automatically dose your plants with a full range of nutrients on a schedule you define.
- **Simple Control:** You can see what's happening on a small screen right on the device. Even better, you can connect to it wirelessly using Bluetooth to change settings, see real-time data, and trigger a manual dose—all from your phone.

### How to Use and Maintain It

This system is built to be simple. Once you've assembled it, the main things you'll do are:

- **Setup:** Use the Bluetooth connection to tell it your target pH, how much nutrient to dose, and how much pH-Up to add in each step.
- **Calibration:** The most important step! To get accurate readings, you'll need to use two common calibration fluids (pH 7 and pH 4) to teach the probe what those values look like. Don't worry, the app makes this super easy.
- **Maintenance:** Keep your pH probe clean. Over time, residue can build up and affect readings, so a quick wipe down every few weeks goes a long way. Make sure your nutrient and pH-Up containers have enough fluid. The system can't dose what isn't there! 😉

### Build Your Own

This project is fully open-source. The core of this system is the software—the brain that controls everything—which is written in C++ and can be uploaded to an Arduino. All the parts are widely available online and are inexpensive. Our goal is to empower you to build, customize, and improve this design. Whether you're a seasoned mycologist, a first-time grower, or an electronics enthusiast, you can replicate this circuit. We encourage you to share your modifications and improvements with the community.

This is more than just a gadget; it's a tool for food sovereignty and a way to share knowledge. We believe that by creating and sharing these tools, we can help build a more resilient and secure food future for everyone. Happy growing! 🍄🌱